

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Consider reaching for a cup. A visual description of its location and a description of the hand's relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identifi-

23 cation with perceptual content is a further modelling task.

24 The mathematical contribution is to specify connections between these
25 requirements and the assumptions each connection needs. A declared ac-
26 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
27 overlaps convert phase order into a bound on content disagreement. Compat-
28 ible spatial measures have a unique global extension; approximate agreement
29 controls a specified global selection. A contracting reconstruction makes that
30 extension its unique fixed point only if it accurately reconstructs the exten-
31 sion. Accuracy across a family of states imposes a further condition on the
32 information retained by the encoding.

33 The empirical contribution is to identify what observations can decide.
34 Stationary phase distributions can be shared by stronger coupling, weaker
35 diffusion and a matched common drive. Finite ramps change the observ-
36 able onset. Coherent adaptation can lose alignment with a designated en-
37 vironmental template, and a compatibility estimator needs diagnostics that
38 distinguish content from decoder artifacts. These findings determine the
39 measurements needed to test the proposed connections.

40 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
41 examines their observables, before Section 6 develops the cortical hypothesis
42 and an awakening experiment. The hypothesis is that extracellular contrac-
43 tion increases field-mediated coupling; its direction, rate and relation to re-
44 covery require independent calibration. Section 7 states the thermodynamic
45 composition and all eight connecting hypotheses per branch. Supporting

models of feedback, learning and resource supply have distinct state spaces
 and assumptions. Supplemental Material organizes their constructions and
 controls behind the claims they support.

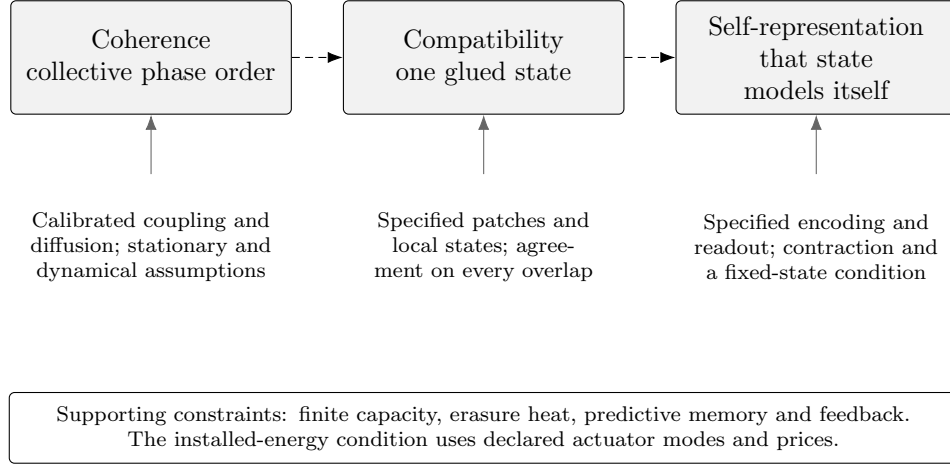


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

49 2 Coherence: physical coupling and its re- 50 source condition

51 2.1 Dynamics and the stationary threshold

52 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

53 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
54 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
55 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
56 parameters. The stationary calculation below uses identical frequencies and
57 this scalar coupling. Applying it to a spatial kernel requires an aggregation
58 rule and a check of the resulting approximation.

59 For identical frequencies in their rotating frame, the positive normalized
60 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

61 where I_n are modified Bessel functions and ψ is the collective phase. The for-
62 mal derivation uses the absence of stationary current for a periodic gradient
63 drift to obtain this density, then verifies normalization and self-consistency.
64 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one

65 positive solution coexists with zero and increases strictly with K . Its onset
 66 is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 Figure 2 shows this stationary branch.

68 Linearization about the uniform density gives rate $K/2 - D$ for the first
 69 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 70 exactly above threshold. About the coherent density, the quadratic form
 71 is strictly dissipative transverse to collective rotation, whose rate is zero.
 72 These linear estimates concern the limiting equation. Relating them to finite
 73 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 74 tion; exchange symmetry alone does not establish the needed independence
 75 (Sznitman, 1991).

76 **2.2 Installed energy bounds mean coupling**

77 A spatial model replaces the finite sum by integration against a kernel. One
 78 declared implementation carries response modes with continuous profiles ϕ_i ,
 79 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

80 The same configuration sets the reciprocal kernel and stored energy. An
 81 executed transition changes the kernel through its changed occupancies and
 82 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The

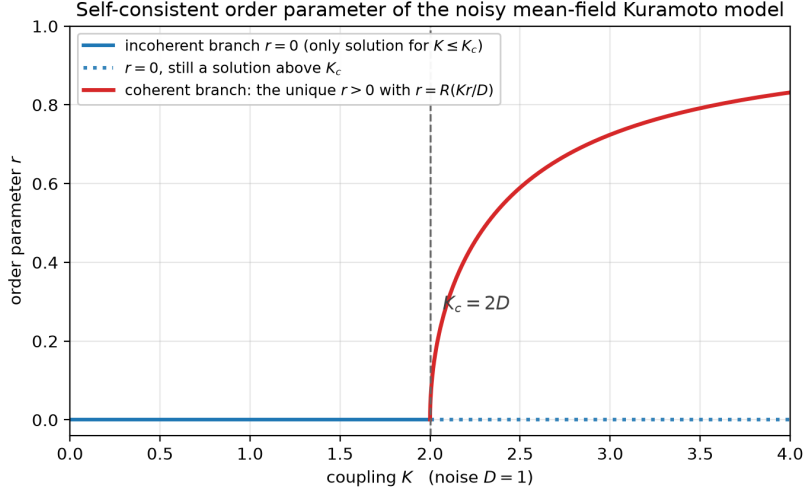


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

83 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 84 pair reconstruction on refining spatial partitions converges to its product-
 85 measure integral under the specified compactness and sampling assumptions;
 86 the refinement index is independent of the number of control updates.

87 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 88 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 89 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

90 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 91 coherent stationary branch and any growing Fourier mode about the uniform
 92 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 93 is necessary: energy above it need not produce sufficient coupling because
 94 the bound can be loose.

95 The quantity constrained is installed energy. Dissipated heat does not
 96 determine it, and κ is supplied by the mode profiles, prices and substrate
 97 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 98 tial dynamics requires a justified reduction. Identifying the installed kernel
 99 with a cortical field is a separate physical commitment.

100 **3 Compatibility: from coordinated activity** 101 **to a global description**

102 **3.1 Why common timing is insufficient**

103 For overlapping patches U and V , compatibility means that restricting either
 104 local description to $U \cap V$ gives the same object. Descriptions can agree on
 105 shared quantities while contributing different information elsewhere. For the
 106 spatial measures used here, compatible restrictions give a unique measure on
 107 the covered domain: the gluing property formalized by a sheaf.

108 In the cup example, visual and reaching descriptions must express a
 109 shared quantity, such as cup location, in common coordinates. Restriction

110 retains that quantity while dropping details particular to each description.
 111 Agreement permits surface appearance and hand configuration to contribute
 112 to one situation. The mathematical counterexample assigns the same phase
 113 to every patch while two local mass profiles disagree on a shared site. Those
 114 profiles have no common extension. Because they are chosen independently
 115 of phase, synchrony places no constraint on their disagreement.

116 This distinction limits the biological interpretation. Receptive territories
 117 and content domains must be related to spatial patches, and the descriptions
 118 must support the required extension. Matching overlap marginals of de-
 119 coded probability laws need not determine a unique global joint distribution
 120 (Abramsky & Brandenburger, 2011). The spatial-measure result supplies a
 121 precise construction; applying it to perceptual content requires a justified
 122 content model.

123 **3.2 When coherence does constrain content**

124 A declared relation between phase and content gives a positive result. Let
 125 contents take values in a real normed vector space E . Patch i reads an
 126 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 127 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 128 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 129 sites; their equality on overlaps is a supplied relation between descriptions of
 130 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 131 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

132 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

133 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 134 agree exactly. The factor N makes this estimate potentially weak in large
 135 populations; it supplies a pointwise guarantee from a population average. A
 136 shared scalar encoder is a special case. Dropping the overlap relation permits
 137 disagreement even at perfect locking. A common nonexpansive linear projec-
 138 tion preserves agreement, but agreement of a projection does not establish
 139 agreement of the full vector.

140 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 141 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 142 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 143 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

144 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 145 duce an initial mismatch. Approximate observations leave a nondecaying
 146 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 147 update to the phase process. These are mechanisms for specified vector con-
 148 tents in shared coordinates; their neural meaning and relevance to experience
 149 remain to be established.

150 **3.3 What approximate agreement determines**

151 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
152 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
153 Weights that vanish outside their patches and sum to one at each site select
154 a global profile by weighted averaging. The selection lies within ε of each
155 patch there, and any two such selections differ by at most ε , independently
156 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
157 $L\varepsilon$ across selections.

158 At zero tolerance, the selection is the unique exact extension. At every
159 positive tolerance there are families with no exact extension; selection then
160 gives an approximation whose dependence on weights is bounded. Supple-
161 mental Material gives a three-site witness attaining the bound. The mass-
162 profile construction uses the scalar case of the content dynamics and addi-
163 tionally requires nonnegativity of the initial profiles and observations. The
164 general vector result supplies no identification with the spatial-measure sheaf.
165 Neither construction selects the neural cover or decoder.

166 4 Self-representation: a state that models it- 167 self

168 4.1 The reflexive construction

169 Compatibility gives a global state. Reflexivity relates it to an encoding within
170 that state. Equip global sections with a complete metric. If the additional
171 restriction-resonance condition identifies encoding with restriction to a region
172 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

173 The region and readout are fixed across the state space. If F contracts and
174 fixes the section s_* formed from the local descriptions, Banach's theorem
175 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
176 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
177 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
178 one for $\tau > 0$. Constructing a map with that factor remains a modelling
179 obligation.

180 In the cup example, the global state includes the represented scene and the
181 perceiver's relation to it. A local encoding must retain enough information
182 for the readout to recover that modelled state; cup position alone would be
183 insufficient if hand configuration and viewpoint could not be reconstructed.
184 Here *prediction* names a reconstruction map. Interpreting it as temporal

185 neural prediction requires dynamics.

186 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 187 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

188 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 189 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 190 weakens. A biological interpretation must specify the represented variables,
 191 metric and acceptable fidelity; changing readouts need a further model.

192 4.2 Reconstruction across a family of states

193 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 194 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 195 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

196 Thus M relevant states pairwise more than 2ε apart require at least M
 197 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 198 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 199 existing self-prediction map.

200 This is an additional accuracy criterion on a family, with no requirement
 201 for multiple exact fixed points. A blind encoder or constant readout fails it
 202 for separated states despite having a unique fixed point. The family, metric

203 and tolerance are inputs; interpreting the code count as physical memory
204 capacity requires a hardware model.

205 **4.3 Agreement after a change takes communication**

206 Snapshot compatibility does not specify how quickly descriptions respond
207 when a shared quantity changes. In a synchronous network whose updates
208 read only the site’s own state and declared neighbours’ messages, two initial
209 configurations agreeing on a site’s causal past through T rounds produce the
210 same state there at round T . If two fresh interventions outside that past
211 require reports separated by more than 2ε , no report from that site can be
212 within ε of both. This follows from the reconstruction bound in Section 4.2.

213 A three-site delay line witnesses failure after one round and exact report-
214 ing after two; cutting the path prevents delivery at every round. A report
215 may still happen to be correct about one fixed world. The bound concerns
216 responsiveness across interventions. Applying it requires a justified commu-
217 nication graph, observation paths and deadline.

218 **5 What the observations establish**

219 The three requirements concern different objects: phase dynamics, local de-
220 scriptions and internal encodings. Their observables must retain these dis-
221 tinctions. The following controls show which mechanisms remain indistin-
222 guishable, which temporal approximations fail, and what independent con-

223 tent measurements must establish.

224 **5.1 Stationary distributions leave the mechanism un-** 225 **determined**

226 Equation (2) gives a concentration–coherence relation with no fitted shape
227 parameter. Estimate r from circular resultant length and a from density
228 shape: inverting R from the measured r would make agreement automatic.
229 Even independently estimated agreement identifies a statistical family shared
230 by different mechanisms.

231 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
232 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
233 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
234 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
235 and diffusion D_0 have that same density. Their stationary distributions co-
236 incide window by window despite different causes.

237 Resolved dynamics supply additional information. In the ideal model with
238 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
239 increment means constrain drift. Separating endogenous coupling from a
240 matched common drive additionally requires a measured drive or controlled
241 perturbation; identifying electromagnetic coupling requires evidence specific
242 to that interaction. Supplemental Material gives the equivalence construction
243 and its numerical check.

244 **5.2 Finite ramps change the observable onset**

245 The square-root law in Eq. (3) describes stationary solutions. A changing
246 coupling requires time for the phase distribution to approach those solutions.
247 In a deterministic mean-field control that crosses the threshold symmetrically,
248 none of the scanned rates meets the declared tracking tolerance: slowing
249 the crossing does not remove the terminal residual under that initialization.
250 A slow control variable therefore supplies no general guarantee of branch
251 tracking at onset. Supplemental Material gives the residual, rate range and
252 admissible rates away from threshold.

253 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
254 at four coupling speeds. Escape is the first of three consecutive saved samples
255 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
256 replicas escape within the fastest window. The three slower ramps have
257 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
258 differ from those of the exact stationary branch evaluated on the same grids.
259 At the fastest speed the ensemble mean never reaches the fit window, so its
260 fitted slope does not estimate stationary onset.

261 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
262 makes the apparent exponent depend on ramp speed and fitting window.
263 The specified deviation-based critical ramp speed is not bracketed by this
264 sweep. With fluctuating coupling, the relevant stationary comparison also
265 depends on rate: the mean coupling approximates order in the tested fast
266 regime, while averaging stationary order over coupling works for the tested

267 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 268 neither approximation. These are finite-horizon observations, with controls
 269 and parameters in Supplemental Material.

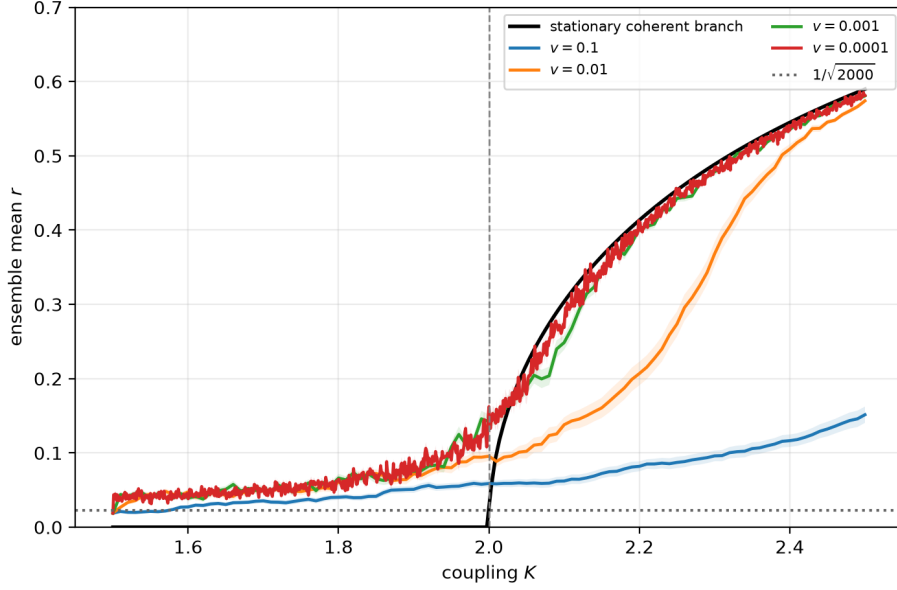


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

270 Additional controls address integration bias, population size and spatial
 271 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 272 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
 273 rameters and controls in Supplemental Material; they supply no cortical cal-
 274 ibration.

275 **5.3 Coherent adaptation can lose environmental align-** 276 **ment**

277 An adaptive coupling rule might supply representational structure, but its
278 objective must reward the relevant agreement. For symmetric coupling, pair-
279 wise sine terms cancel in the population mean drift. Minimizing squared drift
280 at that fixed mean therefore rewards equalizing drift across nodes. In the
281 frequency-cluster control, this favors coupling between unlike-frequency clus-
282 ters, whereas the diagnostic environmental template rewards coupling within
283 clusters.

284 The formalization proves descent of the squared-drift objective for a spec-
285 ified gradient flow at fixed phases. Its functional differs from thermodynamic
286 entropy production: at a coherent stationary density the probability current
287 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
288 rial gives the operator, its gradient and the conditions of the descent result.

289 The finite control co-evolves noisy phases and symmetric, nonnegative
290 couplings at fixed total resource. Its template is a diagnostic covariance of
291 latent signals, never a training input. Across three seeds, second-half order
292 remains between 0.9167 and 0.9222, while the within-cluster to between-
293 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
294 Random-update, relabelled-gradient and frozen-kernel controls are reported
295 in Supplemental Material.

296 The immediate post-update objective reduction disappears when aver-

297 aged over complete update intervals. A prespecified grid of 6 rate–cadence
298 combinations contains no case meeting the 5% sustained-reduction criterion
299 while retaining coherence and losing alignment. The finding is coherent adap-
300 tation away from the specified structure, with no demonstrated sustained cost
301 reduction.

302 Kernel alignment and overlap compatibility are different observables.
303 This control addresses acquisition of its designated template under its par-
304 ticular objective. Learning models can explicitly include data in an objective
305 (Corberi et al., 2026); an account of agreement about the cup must specify
306 the content its learning rule is meant to preserve.

307 **5.4 Content agreement needs independent decoders**

308 A neural test needs independently identified domains, local descriptions and
309 restriction maps. Two decoders should estimate a shared content on overlap-
310 ping domains before their agreement is examined. Discrepancy requires a null
311 under decoding noise and controls using deliberately compatible and incom-
312 patible examples. An implementation on constructed data separates these
313 cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture
314 agreement, and regional bias can manufacture disagreement. Per-patch de-
315 coding diagnostics detect these failures; the compatibility statistic alone does
316 not.

317 Phase agreement cannot replace that measurement. On constructed con-
318 tent, phase carries a share 1.000 of the content entropy when content is built

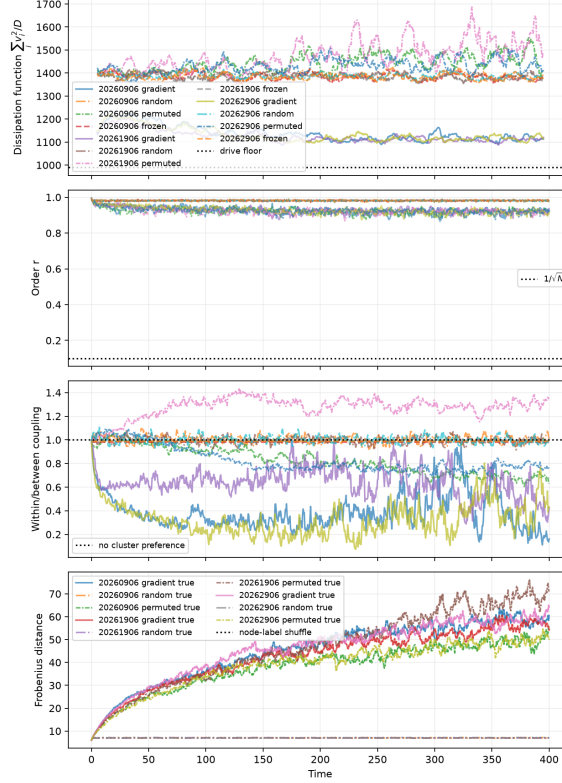


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

319 from it and 0.000 when it is not. A phase-derived statistic reports com-
320 patibility on phase-locked patches whose local descriptions disagree. The
321 implemented content statistic detects their incompatibility. These controls
322 validate a measurement on known constructions; a neural content model and
323 decoder remain to be established.

324 Choosing the cover also chooses a candidate subject: trivial descriptions
325 can agree while carrying little content, and a partition can conceal disagree-
326 ment. The useful choice must therefore be constrained by what neural ac-
327 tivity encodes. The exact composition assumes compatibility of its supplied
328 cover. The content mechanisms above identify additional hypotheses under
329 which agreement can arise; they leave the empirical choice of content and
330 the connection to that cover explicit.

331 **5.5 Phase-density estimation and measurement feasi-** 332 **bility**

333 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
334 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
335 tages produce different residuals. The bipolar results are compatible with the
336 Bessel relation, but their concentration range cannot distinguish it from its
337 linear approximation at the observed scatter. Temporal pooling changes the
338 estimand, and sedation recordings leave the awakening hypothesis untested.

339 The concentration range and dependence determine whether the relation

340 can be tested. Under the specified log-density estimator and bias-floor crite-
 341 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
 342 3000 independent sites in the scanned designs, rising to 31000 when sites
 343 share a mean in groups of 10. These counts describe that estimator and de-
 344 pendence model; the intended recording requires its own calibration. Supple-
 345 mental Material reports the bin-count controls and the full dependence-aware
 346 design.

347 An informative dataset needs sufficient concentration range or precision,
 348 calibrated estimators and an observation model for dependence, source mix-
 349 ing and reference choice. Public mouse calcium imaging provides thousands
 350 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
 351 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
 352 here. Site count, state coverage and bandwidth are separate requirements.

353 **6 The cortical hypothesis and an awakening** 354 **test**

355 The cortical proposal identifies a physical source of coupling and asks whether
 356 its changes help explain recovery. The tests in Section 5 determine what
 357 observations must distinguish before a phase trace can support this identifi-
 358 cation.

359 6.1 Physical identification and calibration

360 The cortical hypothesis identifies part of effective coupling with endogenous
361 extracellular electromagnetic interactions. Fields can modulate spike timing
362 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
363 propose coordination of distributed neural information (Pinotsis & Miller,
364 2022; McFadden, 2020). Phase organization under propofol motivates study-
365 ing this regime (Bastos et al., 2021), but does not measure coupling.

366 The formal field predicate requires a jointly continuous kernel, a normal-
367 ized substrate, coupling equal to the kernel average, and no single site pro-
368 viding more than half that strength. Toy constructions satisfy it. A cortical
369 realization must specify the scale at which the kernel and its reduction ap-
370 ply: calcium imaging reports different state-dependent modularity patterns
371 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

372 The spatial reduction is also testable. A row-sum aggregation at zero
373 diagonal tracks the uniform-coupling control over the tested spatial decay
374 range, while frequency heterogeneity displaces the operational threshold. The
375 fit window biases both estimates relative to $2D$; agreement with the control
376 is therefore the supported conclusion. Supplemental Material reports the
377 reduction, its numerical error and the distinct assumptions needed for the
378 kernel-average identification in Eq. (5).

379 The field commitment requires physical evidence beyond a choice of math-
380 ematical representation. A finite matrix spread over positive-mass cells pre-
381 serves its total coupling and phase correlation exactly. Point-supported

382 vanishing under an atomless measure and a disadvantage from excluding
 383 a better-correlated pair concern specified support constraints. Neither com-
 384 parison supplies energy prices or a biological–digital distinction.

385 A spike-time shift alone supplies no coupling rate. The dimensionless
 386 product of population size N , timing sensitivity s , field amplitude E and
 387 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (11)$$

388 This rate describes how induced timing shifts accumulate into drift. Nei-
 389 ther it nor the installation factor κ follows from a heat bound. Testing the
 390 physical threshold requires independent estimates of the interaction kernel,
 391 its aggregation and diffusion in the same cortical regime.

392 **6.2 Extracellular geometry and recovery**

393 The proposed control variable is extracellular geometry. Extracellular vol-
 394 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 395 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 396 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 397 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 398 an effective-medium description predicts that a larger extracellular fraction
 399 raises conductivity and reduces field amplitude.

400 The coupling-gated hypothesis consequently predicts that extracellular

401 contraction on awakening raises E and, holding the other factors in Eq. (11)
 402 fixed, raises effective coupling. This argument supplies a direction; the
 403 geometry–coupling relation, its magnitude and its role in human recovery
 404 require measurement.

405 For a smooth crossing at nonzero speed, evaluating the stationary branch
 406 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (12)$$

407 This follows from Eq. (3) at fixed D and concerns the stationary branch
 408 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 409 dently justified tracking approximation. The crossing controls in Section 5.2
 410 leave that approximation unsupported at onset, even though a separation of
 411 geometry and phase time scales can support it away from threshold. The
 412 awakening hypothesis therefore needs a calibrated dynamical prediction for
 413 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 414 does not supply a recovery latency.

415 **6.3 A protocol for the awakening experiment**

416 The physical test has six requirements:

- 417 1. **Coupling and diffusion in the same units.** Estimate them from re-
 418 solved phase dynamics, separating heterogeneity and observation noise.
 419 Measure the drive or perturb the system to distinguish endogenous cou-

420 pling.

421 **2. Geometry calibrated to coupling.** Validate the geometry observ-
422 able and its relation to effective coupling, with uncertainty. The con-
423 ductivity argument supplies only the expected direction.

424 **3. Calibrated phase measurements.** Estimate r by resultant length
425 and a by unweighted regression of log density on centered cosine.
426 Choose site count and binning from a dependence-aware calibration
427 over the intended concentration range. The tested designs require
428 thousands of sites at the observed concentrations; temporal pooling
429 changes the spatial estimand.

430 **4. An independent recovery endpoint.** Measure behavioral recovery
431 separately from phase organization.

432 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
433 old above the finite-size floor and evaluate its latency under the mea-
434 sured coupling trajectory, including ramp speed or fluctuations.

435 **6. Matched model comparison.** Compare coupling increase, diffusion
436 change, common drive and alternative recovery models under the same
437 observation and filtering assumptions.

438 Failure of the concentration–coherence relation over an informative range
439 constrains the density model. Failure of the calibrated geometry–coupling
440 association constrains the physical gate. Better prediction of independent

recovery by a competing mechanism counts against its proposed role. A favorable outcome still leaves the content of coordinated activity to be explained.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Laudauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still et al., 2012). Zero retained information also satisfies it, so a task and a learning rule require additional specification. Relating this bound to Lan-

459 dauer heat requires a common thermal scale and budget, and memory waste
 460 bounded by erased entropy.

461 An acting agent changes the signal it will receive. Specify policy, world,
 462 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{14}$$

463 The second conditional term vanishes in the passive model. With feedback,
 464 the signed difference can be negative as action creates correlations with the
 465 future. A causal reading requires an environmental state sufficient to describe
 466 the relevant dynamics.

467 For a controlled substep, the active thermodynamic bound instead uses
 468 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 469 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 470 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 471 2013). Every forward path of positive probability must have a positive-
 472 probability reverse for this finite identity. A missing reverse gives infinite
 473 divergence. The supplement develops the support conditions and composes
 474 actuation with memory updates on their shared intermediate law.

475 **7.2 Learning and repeated operation need explicit re-** 476 **sources**

477 The finite examples distinguish selecting a feasible policy, relaxing a register
478 toward an encoded reward, and learning an unknown task from observa-
479 tions. The observational learner acquires correlation with an environmental
480 parameter while its action marginal stays uniform. A continuing model exe-
481 cutes actuation, observation-driven learning and reset in sequence, preserving
482 above-chance performance at a positive work cost per cycle.

483 A phase-derived sensor makes the informational distinction concrete: an
484 informative readout and a blind control have the same phase order but dif-
485 ferent learning performance. Attaching the learner’s register to the actuator
486 of Section 2.2 ties reward, installed kernel and work to its own executed law.
487 The learning witness rises above the necessary installed-energy bound while
488 its matched control remains on the exclusion line. These finite constructions,
489 detailed in Supplemental Material, leave the evolving feedback between phase
490 dynamics, installation and sensing to be specified.

491 For that model, common energy E , thermal scale θ , local detailed balance
492 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (15)$$

493 where expected cumulative work stays within the supplied allowance B at
494 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

Table 1 (continued)

Connection	Status	Required assumption
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Joint witnesses combine a register, boundary, refining mesh and small spatial cover; active witnesses add controlled thermal updates and a priced installation. They establish satisfiability under their stated identifications. Mesh convergence follows from refinement data, and the witnesses establish no common cortical mechanism. Supplemental Material maps the claims to their Lean declarations and records the counterexamples. The development declares no additional axioms beyond its libraries’ foundational axioms.

526 8 Discussion

527 8.1 Unity and a minimal self as empirical commit- 528 ments

529 We propose the state glued from compatible local descriptions as a corre-
530 late of conscious unity, and its accurate internal reconstruction as a correlate
531 of a minimal self. The motivations are that different contents belong to
532 one experienced situation and that the situation is experienced from a per-
533 spective. The conditional results identify relations a realization would need:
534 sufficient installed resources for the specified phase model, agreement about
535 shared content, and an encoding that supports reconstruction at the required
536 fidelity.

537 Compatibility allows different descriptions to contribute different infor-
538 mation. It also permits uncertainty and conflicting judgments: apparent cup
539 position and judged cup position can be distinct quantities in a consistent
540 description of their discrepancy. Which quantities are shared must be de-
541 termined independently. Choosing them solely to secure agreement would
542 deprive the proposal of empirical force.

543 The minimal-self proposal concerns the perceiver’s position and condition
544 within the situation, such as the cup’s location relative to a body that can
545 reach it. The encoding could be implicit and need carry no autobiograph-
546 ical narrative. Its proposed necessity is philosophical: even an unreflective
547 episode has a perspective. Gluing by itself requires no self-encoding, and re-

constructability alone supplies no functional role. Altering the encoded hand or viewpoint while holding the external scene fixed should change estimates of the perceiver’s relation to the cup as the readout predicts. Whether that represented perspective is experienced remains a further question.

8.2 What a test of the full proposal must establish

Both uniqueness results concern specified objects. They choose neither the cover nor the encoding, and do not select between candidate subjects. Integrated information theory treats exclusion explicitly (Tononi et al., 2016); here selection requires independent evidence about neural organization. A constant readout can contract and fix a predetermined state while discarding its input. The family criterion in Section 4.2 tests the sensitivity that this construction lacks, provided the relevant states and fidelity are declared independently.

A test must separately measure coupling relative to diffusion, content compatibility and self-reconstruction, alongside an independent endpoint for experience. Meeting all three without a unified episode would challenge sufficiency. A unified episode with incompatible descriptions would challenge the unity identification; one lacking the specified self-encoding would challenge the reflexive requirement. Awakening, dreaming and dissociative states offer possible settings, but absence of report cannot by itself define absence of experience.

The next physical requirement is a calibrated geometry–coupling relation

570 and resolved phase dynamics. The installation bound contributes a nec-
571 essary resource condition where its hardware representation applies; cortical
572 mode prices and field-mediated coupling require measurement. For the repre-
573 sentational connection, an informative experiment asks whether a controlled
574 change in coordination alters independently decoded content agreement, with
575 controls for sensory information and decoder quality. The constructed-data
576 estimator establishes feasibility for specified contents. Neural local descrip-
577 tions, shared overlap encoders and the self-encoding have yet to be identified.
578 The population-size dependence of the phase-to-content bound further limits
579 its quantitative use.

580 The separation results explain why these measurements must be inde-
581 pendent. A stationary distribution leaves several recovery mechanisms indis-
582 tinguishable. Equal phase order can accompany incompatible descriptions
583 or different task information. The tested plasticity rule preserves coherence
584 while losing its designated template. Each finding concerns the specified
585 model and observable; their role is to delimit the evidence needed for the
586 proposed connections.

587 **8.3 Explanatory scope**

588 Integrated-information, global-workspace and predictive-processing accounts
589 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
590 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
591 candidate physical interaction and conditional relations between local content

592 and self-encoding. Gluing and reconstruction permit multiple substrates;
593 the electromagnetic commitment enters through physical calibration. The
594 formalization checks implications under their hypotheses, and its witnesses
595 establish realizability on declared models. They do not together supply a
596 cortical agent.

597 The explanatory question is whether the specified relations occur in neural
598 activity and matter for experience. In the cup example, this means establish-
599 ing how coordination helps maintain one represented situation and how the
600 perceiver’s place within it contributes to the episode. The proposal identi-
601 fies candidate correlates and tests of that identification. Why those relations
602 would be accompanied by experience remains unexplained.

603 **Data and code availability**

604 The Lean development, simulation scripts, tests and saved numerical sum-
605 maries are maintained in the manuscript source repository. Publication
606 simulation values are generated from saved summaries without rerunning
607 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
608 ds005620 (Bajwa et al., 2025); preprocessing and selection are described in
609 Supplemental Material. The repository contains the reproducible analysis
610 and does not redistribute the raw recordings.

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614 **Use of AI tools**

615 AI assistance to this work was substantial and is not separable by task.
616 Several coding assistants and chat interfaces were used in an interleaved way
617 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
618 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
619 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
620 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
621 they contributed to the Lean development, the simulation and analysis code,
622 and the drafting and revision of both documents. Because the tools were
623 used in combination and in alternation, no section, proof, figure or result is
624 attributable to any one of them, and none is claimed to be. The author is
625 responsible for all content, including everything a tool produced. Machine
626 checking establishes the stated Lean results under their hypotheses; it does
627 not validate biological interpretations or replace scientific review of generated
628 code and prose.

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